BDM Session 4

Validated algorithm implementation, ablation, and catalog migration

Cheng-Zong Ruan



Institute for Computational Cosmology, Durham University

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Executive finding: validated mode is not ready

NO-GO: `validated_v1`

The integrated v1.7 candidate stopped before the mandatory N128 cosmological catalog could be serialized.

- 126 surviving candidates had undefined M_{200c};
- 9 were at or above the frozen 500-particle gate;
- reciprocal matching and frozen HOD/HSM propagation were unavailable.

RETAIN: `legacy_corrected`

Production-compatible default with Session 3 safety fixes, unchanged legacy physics, and unchanged `Catshort` semantics.

175/175 final login-safe and real-Fortran tests pass

0 thread/repeat representation floor

24 columns: `Catshort_legacy_v0`

Evidence: `results/bdm_validated_algorithm/manifest.json`

Three execution modes, three different contracts

legacy_original

Archived historical executable only.

Exact reproduction; unsafe behavior is never reintroduced into current source.

`PMP2BDM.exe`
`c84f4e3c…ad89`

legacy_corrected

Default after Session 3.

Safety fixes only; legacy physics, defaults, filenames, and 24-column schema stay compatible.

Fast, debug, and strict-reference executables frozen separately.

validated_v1

Explicit opt-in candidate.

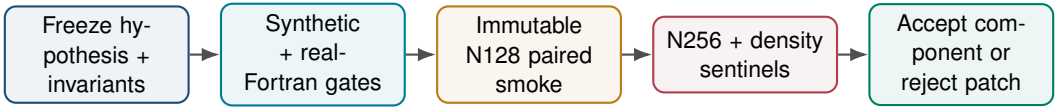
Separate filenames, versioned fields, and fail-closed metadata.

`BDMMode=validated_v1` experiments require an explicit environment guard.

A component can pass its own definition tests without making the integrated validated mode production-ready.

Evidence: Mode/configuration patch 0001; baseline freeze

Evidence gate for every behavior change



9 × 49 recorded synthetic finder campaigns

54 legacy identity comparisons

46 machine-readable patch comparisons

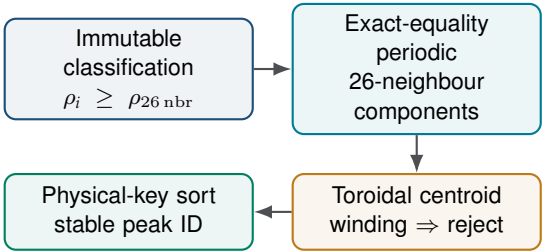
38 finder runs in 12 Slurm jobs

Reject on: non-finite values, bounds/FPE failures, lost legacy identity, non-zero invariance, SO closure failure, or unresolved physics.

No thresholds were loosened after inspecting a failed result.

Evidence: Per-patch pre/post JSON and `ablation.csv`

Deterministic exact-plateau construction



Sort key: peak density ↓; toroidal (x, y, z) ↑; plateau cell count ↓; second moment ↑. An identical physical key fails diagnostically instead of falling back to array order.

49/49 metamorphic finder runs pass

210 metric comparisons at zero floor

119 / 1303 N128 / N256 reciprocal matches

Every matched legacy field was identical. No multi-cell plateau survived into either cosmological catalog.

Evidence: Patch 0002: accepted plateau component

One radial profile, unambiguous mass semantics

Legacy quantities remain explicit

$M_{\rm legacy}, R_{\rm legacy}$	historical all-particle extended aperture; not SO
$M_{\rm bound}, N_{\rm bound}$	historical one-pass Newtonian result; exact count exposed

All-particle SO quantities

$(M_{\rm vir}, R_{\rm vir}, N_{\rm vir})$,
$(M_{\rm 200m}, R_{\rm 200m}, N_{\rm 200m})$,
$(M_{\rm 200c}, R_{\rm 200c}, N_{\rm 200c})$.

$$M_\Delta = \frac{4\pi}{3}\,\Delta\rho_{\rm ref}\,R_\Delta^3$$

One sorted minimum-image profile; largest discrete enclosed count above target density; analytic paired radius. The internal particle bracket is enforced, but the public closure envelope is $M_\Delta \pm m_{\rm p}/2$.

Status 2 means **no discrete crossing**. Its exact zero is a sentinel, never a zero-mass measurement. Readers must fail rather than drop it.

Evidence: Patches 0003–0003d: common SO profile and closure semantics

SO completeness is the hard blocker

Candidate	State	undefined	≥ 500	catalog?	exit	Decision
v1.2.1	N128	57	0	no	128	reject zero-unresolved policy
v1.3	N128	57	0	yes	0	snapshot passes
v1.3	N256	44	3	no	128	reject integrated catalog
v1.7	N128	126	9	no	128	mandatory smoke fails
v1.7	N256	77	21	no	128	inherited and worsened gate

The v1.2 half-particle envelope itself is translation-covariant: 1,092 SO-field comparisons were exact. The failure is **profile completeness for retained rows**, not closure arithmetic.

9 / 21 N128 / N256 failures above the 500-particle gate

Without a catalog: no reciprocal matching, HMF/exclusion/ξ_{hh}, velocity/environment, or frozen HOD/HSM propagation.

Evidence: Patches 0003c, 0003d, and 0007

Centre, velocity, unbinding: still provisional

Tolerance recenter: rejected

Periodic fixed-aperture all-particle centre of mass; binary64 stable accumulation; explicit tolerance; maximum eight passes.

One frozen 10,000-particle NFW phase did not reach $10^{-4}\,h^{-1}\mathrm{Mpc}$. The tolerance was not widened.

Alternatives: validation-only

- central-core, whole-aperture, and central-particle velocities;
- additional iterative unbinding passes;
- conventional all-particle SO without unbinding.

Candidate metadata therefore states the actual retained definitions: legacy extended-aperture bulk velocity and legacy one-pass Newtonian unbinding. No f(R), CPL, or generic-MG bound mass is recommended.

No centre, velocity, or unbinding alternative became a production default.

Evidence: Patch 0004 and deferred validation controls

Internal-property kernels improved selectively

Global resolved $V_{\max}$: component accepted

Maximize $GM_{\mathrm{all}}(<r)/(ar)$ globally; require at least 20 enclosed particles; largest-radius tie-break. Report raw value, $R_{\max}$, exact count, status, and a separate non-default correction diagnostic.

N128: 119 reciprocal matches; all pre-existing fields and downstream statistics exact. Raw/global versus historical corrected $V_{\max}$: median -1.39%, p95 1.74%, tail 2.01%.

Symmetric eigensolver: component accepted

Scale-normalized binary64 cyclic Jacobi; deterministic eigenvalue order, vector signs, and degeneracy status. Trusted tests cover 263 analytic, random, degenerate, and nearly singular matrices.

Raw reduced-shape schema rejected: rank-deficient axis-ratio errors exceed the frozen 5×10^{-13} bound, and periodic translations are non-zero.

Numerical kernels accepted; empirical corrections and convergence claims are not.

Evidence: Patches 0005 and 0006

Five distinct periodic-classification concepts

Numerical duplicate
exact centre + M_{legacy} + R_{legacy} + N_{all}; smallest stable peak ID wins.

Inside a larger host
periodic centre lies within a strictly larger host's R_{legacy}.

Distinct HSM object
neither a numerical duplicate nor a subhalo; mass remains M_{legacy}.

Subhalo
inside primary host and $M_{\mathrm{cand}}/M_{\mathrm{host}} < 2/3$.

Major-merger ambiguity
contained pair with mass ratio $\geq 2/3$; velocity is not used.

Primary host: largest mass, then nearest minimum-image distance, then smallest stable peak ID.

42 classification metamorphic comparisons

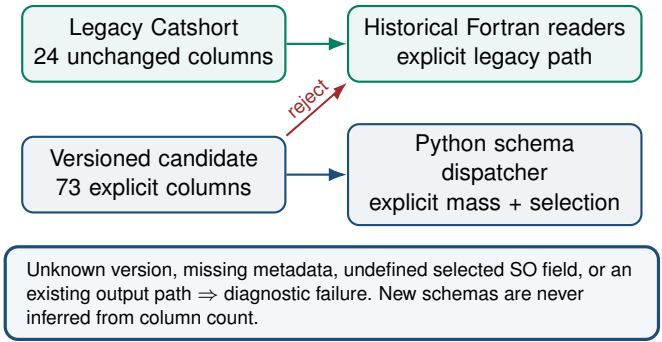
0 payload or stable-ID differences

49/49 integration campaign generic gates

Kernel accepted; v1.7 candidate-retention integration rejected because retained rows lack complete SO profiles.

Evidence: Patch 0007

Catalog migration now fails closed



Downstream contract

HMF/HSM production stays on column-8 $M_{\rm legacy}$ and the historical population. $M_{\rm 200m}$ is not an HSM default.

Honest approximation labels

`M500c_NFW_legacy_approx`
`M200c_NFW_legacy_approx`

Neither is a measured SO mass.

Evidence: Patch 0008; catalog schema and migration notes

Reproducible validation with modest resources

175/175 final tests across fast, debug, strict-reference

12 shared `cosma8-serial` Slurm jobs

1017.325 s recorded TotalCPU

0 canonical thread/repeat numerical floor

38 cosmological finder invocations

1.93 GiB maximum batch MaxRSS

46 × 39 ablation rows × fields

0.35694 billed core-hours

0 new simulations or full-node jobs

Eight root jobs are recorded as failed because hard-gate failures were deliberately propagated and preserved. No failure output, catalog, snapshot, manifest, or historical executable was overwritten.

Evidence: Final manifest and Session 4 Slurm accounting

Patch outcomes by algorithm area

Area	Accepted for future composition	Rejected or deferred
Peaks	periodic exact-equality plateau components; stable physical IDs	array-index tie breaks; tolerance merging
SO profiles	common profile kernel; half-particle closure envelope; explicit undefined status	raw public brackets; incomplete integrated catalog
Centre/velocity	explicit status and metadata	tested tolerance recenter; alternate velocity defaults
Unbinding	explicit legacy one-pass meaning	iterative production default; unresolved MG bound mass
V_{max}	global resolved maximum, radius, count, raw/diagnostic split	resolution correction as a default
Shape	cyclic-Jacobi symmetric eigensolver	raw reduced-shape schema; empirical correction
Classification	deterministic periodic duplicate/host/subhalo/merger kernel	retain-all v1.7 integration
Catalogs	fail-closed version dispatch; explicit legacy approximation labels	schema inference; silent row dropping; $M_{200\mathrm{m}}$ HSM default

Well-tested components and explicit semantics are now available, while production remains at the last fully reproducible baseline.

Evidence: Patch decisions 0001–0008

Production recommendation stays conservative

Use now

- mode: `legacy_corrected`;
- schema: `Catshort_legacy_v0`;
- HMF/HSM mass: $M_{\rm legacy}$;
- historical distinct/sub selection;
- frozen Session 3 configuration and checksums.

Do not infer

- no production particle or force-mesh resolution;
- no converged HMF/clustering/velocity/property threshold;
- 500 particles is a validation floor, not a recommendation;
- no MG bound-mass prescription;
- no relabeling of existing emulators as SO mass.

Next milestone: complete versioned N128/N256 catalogs; then reciprocal HMF, exclusion, ξ_{hh}, velocity/environment, and frozen HOD/HSM propagation without refitting or threshold tuning.

Evidence: `recommended_configuration_lcdm.json`

Take-away

NO-GO for integrated `validated_v1`

Complete SO semantics for every retained row before adoption.

Keep `legacy_corrected` in production

Reproducible, byte-compatible, and fully gated.

Deterministic plateaus | explicit SO closure/status | global $V_{\max}$ | trusted eigensolver | periodic classification | fail-closed catalogs

Evidence: Session 4 final decision, commit `513b4a1`